

Path Integral.

Sec. 6th

(1)

Consider a non-relativistic QM in one dimension.

$$H(P, Q) = \frac{1}{2m} P^2 + V(Q).$$

P and Q are operators. $[Q, P] = i\hbar$.

We want to evaluate the particle transition probability.

Starts at q' at t' and end at q'', t''

$$\langle q'' | e^{-iH(t''-t')} e^{iHt'} | q' \rangle$$

where $|q''\rangle$ and $|q'\rangle$ are eigenstates of Q .

as the time evolution of $|q'\rangle$ is e^{-iHt}

Formulating this question in the Heisenberg picture

$$Q(t) = e^{-iHt} Q(0) e^{iHt}$$

1 Not needed until p. 6: pp. 2-5 use only $\langle q'' | e^{-iH(t''-t')} | q' \rangle$. $Q(t) = e^{iHt} Q e^{-iHt}$ and $|q, t\rangle = e^{iHt} |q\rangle$ are used from p. 6 on, for the time-ordered insertions that appear in LSZ.

The instantaneous eigenstate of $Q(t)$ can be defined via

$$Q(t) |q, t\rangle = q |q, t\rangle$$

$$\text{and } |q, t\rangle = e^{+iHt} |q\rangle$$

$$\text{where } Q |q\rangle = q |q\rangle$$

↑
t=0.

(2)

Then the transition amplitude is as simple as

$$\langle q'', t'' | q', t' \rangle \text{ in the Heisenberg picture.}$$

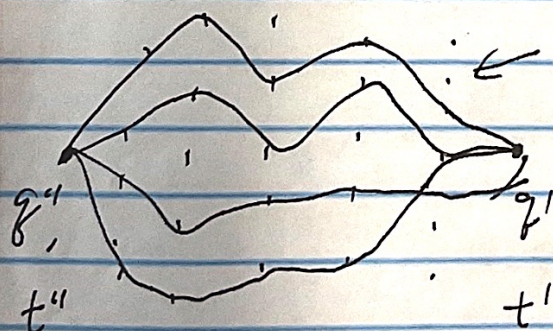
To evaluate, we can do the following.

$$\langle q'', t'' | q', t' \rangle = \int \prod_{j=1}^N dq_j \langle q'' | e^{-iH\delta t} | q_N \rangle \langle q_N | e^{-iH\delta t} | q_{N-1} \rangle \dots \langle q_1 | e^{-iH\delta t} | q' \rangle$$

by inserting N complete states. $|q_i\rangle\langle q_i| \int_{-\infty}^{\infty} dq_i$

basically I am evaluating the transition as a product of small ~~the~~ transitions

$$\delta t = \frac{t'' - t'}{N} \quad (1)$$



← each path has different weight (2)

1 N insertions give $N + 1$ factors of $e^{-iH\delta t}$, so $\delta t = (t'' - t')/(N + 1)$, matching the $N + 1$ momentum integrals on p. 4. Irrelevant as $N \rightarrow \infty$.

2 Every path has weight $e^{iS[q]}$ of modulus one: paths differ in phase, not size. Near the classical path S is stationary and neighbours add coherently; elsewhere they cancel. Real damping enters only through $(1 - i\epsilon)H$, p. 8.

(3)

1 Sign: $e^{-iH\delta t} - e^{-iA\delta t}e^{-iB\delta t} = +\frac{1}{2}[A, B]\delta t^2 + \dots$, because the BCH exponents here are $-iA\delta t$ and $-iB\delta t$, and $(-i)^2 = -1$. The accumulated error $\sim t\delta t \rightarrow 0$ is unchanged.

For any small amplitude considered, using the Campbell - Baker - Hausdorff formula

$$\exp(A+B) = \exp A \exp B \exp(-\frac{1}{2}[A, B] + \dots)$$

$$e^{-iH\delta t} = e^{-i\frac{p^2}{2m}\delta t} e^{-iV(q)\delta t} [O(\delta t^2)] \quad \text{additional terms if}$$

The error I am making is

$$e^{-iH\delta t} - e^{-iA\delta t}e^{-iB\delta t} = -\frac{[A, B]}{2}\delta t^2 + \dots \quad \text{upon integration, } -\frac{[A, B]}{2}\delta t \xrightarrow{\text{fast limit}} 0 \quad \text{if } [A, [A, B]] \neq 0 \text{ or } [B, [A, B]] \neq 0 \quad (1)$$

for small δt ~~and other scales of this order~~

we approximately have

$$\begin{aligned} \langle q_2 | e^{-iH\delta t} | q_1 \rangle &= \int dp_1 \langle q_2 | e^{-i\frac{p_1^2}{2m}\delta t} | p_1 \rangle \langle p_1 | e^{-iV(q_1)\delta t} | q_1 \rangle \\ &= \int dp_1 e^{-i\frac{p_1^2}{2m}\delta t} e^{-iV(q_1)\delta t} \underbrace{\langle q_2 | p_1 \rangle \langle p_1 | q_1 \rangle}_{\frac{1}{(2\pi)^{1/2}} e^{-ip_1 q_1}} \end{aligned}$$

$$= \int dp_1 e^{-i\frac{p_1^2}{2m}\delta t} e^{-iV(q_1)\delta t} \frac{1}{(2\pi)} e^{-i(p_1 q_1 - p_1(q_1 - q_2))}$$

$$= \int \frac{dp_1}{(2\pi)} e^{-iH(p_1, q_1)\delta t} e^{-ip_1(q_1 - q_2)}$$

Trotter Product Formula:

$$\lim_{N \rightarrow \infty} (e^{-iA\epsilon} e^{-iB\epsilon})^N = e^{-i(A+B)t}$$

$t \equiv N\epsilon$.

1 $\bar{q}$ is the average of the slice's two endpoints. For $P^2/2m + V(Q)$ the endpoint form of p. 3 has the same $\delta t \rightarrow 0$ limit; the midpoint matters when H mixes P and Q . For Weyl-ordered H , $\langle q_2 | H | q_1 \rangle = \int \frac{dp}{2\pi} H(p, \bar{q}) e^{ip(q_2 - q_1)}$ exactly.

(4)

Here $H(p, q)$ is simple and we could do separation of variables, but. for more complex ones, e.g. $V(Q) \rightarrow V(p, q)$. We can adopt

Weyl ordering ← from classical hamiltonian $H(p, q)$

$$H(p, q) \equiv \int \frac{dx}{2\pi} \frac{dk}{2\pi} e^{-ixP + ikQ} \int dp dq e^{-ixp - ikq} H(p, q) \delta(p - p) \delta(q - q)$$

Then the corrected transitions amplitude would be^①

$$\langle q_2 | e^{-iH\delta t} | q_1 \rangle = \int \frac{dp_1}{2\pi} e^{-iH(p_1, \frac{1}{2}(q_1 + q_2))\delta t} e^{ip_1(q_2 - q_1)}$$

Adopting Weyl ordering for the general case:

$$\langle q'', t'' | q', t' \rangle = \int \prod_{k=1}^N dq_k \prod_{j=0}^N \frac{dp_j}{2\pi} e^{ip_j(q_{j+1} - q_j)} e^{-iH(p_j, \bar{q}_j)\delta t}$$

isn't averaging over initial & final more satisfying?
 $\bar{q}_j = \frac{1}{2}(q_j + q_{j+1})$ - $q_0 = q'$, $q_{N+1} = q''$
 If we now define $\dot{q}_j = \frac{q_{j+1} - q_j}{\delta t}$ and take the limit $\delta t \rightarrow 0$, we get

$$\langle q'', t'' | q', t' \rangle = \int Dq Dp \exp \left[i \int_{t'}^{t''} dt (p(t)\dot{q}(t) - H(p(t), q(t))) \right] \quad \dots (6.8)$$

One can further simplify by noticing that for simple enough $H(p, q)$

(p less than p^2 , and p^2 has not q dependence), or equal to

1 Each p_j integral is Fresnel, not a real Gaussian:
 $\int \frac{dp}{2\pi} e^{i\delta t(pq - p^2/2m)} =$
 $(\frac{m}{2\pi i \delta t})^{1/2} e^{i\delta t m q^2/2}$, the box continued to imaginary width; $(1 - i\epsilon)H$ (p. 8) makes it converge. The prefactor is q -independent and goes into the new Dq .

The E.O.M is the replacement equation for p to finish the Gaussian integral.

(this is Gaussian and the normalization / prefactor)

$$\langle q'', t'' | q', t' \rangle = \int Dq \exp. \left[i \int_{t'}^{t''} dt L(\dot{q}(t), q(t)) \right] \quad (1)$$

$L(\dot{q}, q)$ is computed by first finding the stationary point of the p integral by solving

$$0 = \frac{\partial}{\partial p} (p\dot{q} - H(p, q)) = \dot{q} - \frac{\partial H(p, q)}{\partial p}$$

for p in terms of $\dot{q}$ and q , and then plugging back into $p\dot{q} - H$ to get L .

Gaussian integral

$$\int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}$$

$$\rightarrow \int_{-\infty}^{+\infty} e^{-(x+a)^2} d(x+a) = \sqrt{\pi}$$

Quick derivation

$$I = \int_{-\infty}^{+\infty} e^{-x^2} dx$$

$$I^2 = \int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} dy e^{-(x^2+y^2)}$$

$$= \int_{-\infty}^{+\infty} \int_0^{2\pi} r dr d\theta e^{-r^2}$$

$$= \int_0^{2\pi} d\theta \int_0^{\infty} r dr e^{-r^2}$$

$$= 2\pi \cdot \frac{1}{2} \int_0^{\infty} dr^2 e^{-r^2} = -\pi [e^{-r^2}]_0^{\infty} = \pi$$

$$\int_{-\infty}^{+\infty} e^{-(x/a)^2} dx = \frac{\sqrt{\pi}}{|a|} \cdot a \sqrt{\pi}$$

(6)

Now let's try to generalize:

$$\langle q'', t'' | Q(t_1) | q', t' \rangle = \langle q'' | e^{-iH(t'' - t_1)} Q e^{-iH(t_1 - t')} | q' \rangle$$

In the path-integral formula.

this $Q(t_1)$ will generate an extra $q(t_1)$

[e.g. from 6.7] $\langle q'', t'' | q', t' \rangle = \int \pi dq_k \pi \frac{dp_j}{2\pi} e^{i p_j (q_{j+1} - q_j) - i H \delta t}$

$$\langle q'', t'' | Q(t_1) | q', t' \rangle = \int \mathcal{D}p \mathcal{D}q q(t_1) e^{iS}$$

where $S = \int_{t'}^{t''} dt (p\dot{q} - H)$

Now consider the opposite side $\int \mathcal{D}p \mathcal{D}q q(t_1) q(t_2) e^{iS}$

this clearly requires time ordering $Q(t_1)$ and $Q(t_2)$

if $t_1 < t_2$ $Q(t_2) Q(t_1)$

if $t_1 > t_2$ $Q(t_1) Q(t_2)$

so $\int \mathcal{D}p \mathcal{D}q q(t_1) q(t_2) e^{iS}$

$$= \langle q'', t'' | T Q(t_1) Q(t_2) | q', t' \rangle$$

$\uparrow$
this is what appears in LSZ.

We can quickly realize that we are dealing with functions.

⑦

We can define the functional derivative
variation

$$\frac{\delta f(t_2)}{\delta f(t_1)} = \delta(t_2 - t_1)$$

(it satisfies the product rule, chain rule &c.)

Now, consider modifications to the Lagrangian.

$$H(p, q) \rightarrow H(p, q) - f(t)g(t) + h(t)p(t)$$

$f(t)$ and $h(t)$ are specified functions

$$\langle q'', t'' | q', t' \rangle_{f,h} = \int \mathcal{D}p \mathcal{D}q e^{i \int_{t'}^{t''} dt (p\dot{q} - H + f g + h p)}$$

1 Each derivative needs its own $1/i: \frac{1}{i} \frac{\delta}{\delta f(t_1)} \frac{1}{i} \frac{\delta}{\delta f(t_2)} \langle \rangle_{f,h} = \int \mathcal{D}p \mathcal{D}q q(t_1) q(t_2) e^{iS}$. The same holds for every derivative in the last line of this page.

then

$$\frac{1}{i} \frac{\delta}{\delta f(t_1)} \langle q'', t'' | q', t' \rangle_{f,h} = \int \mathcal{D}p \mathcal{D}q g(t_1) e^{iS_{f,h}}$$

$$\frac{1}{i} \frac{\delta}{\delta f(t_1)} \frac{\delta}{\delta f(t_2)} \langle \quad \quad \rangle_{f,h} = \int \mathcal{D}p \mathcal{D}q g(t_1) g(t_2) e^{iS_{f,h}} \quad (1)$$

$$\frac{1}{i} \frac{\delta}{\delta h(t_1)} \langle \quad \quad \rangle_{f,h} = \int \mathcal{D}p \mathcal{D}q p(t_1) e^{iS_{f,h}}$$

so we can get any # of p, q and then set

$h(t) = f(t) = 0$. that get us back to the original Lagrangian.

$$\langle q'', t'' | T(Q(t_1) \dots P(t_n) \dots | q', t' \rangle$$

$$= \frac{1}{i} \frac{\delta}{\delta f(t_1)} \dots \frac{\delta}{\delta f(t_n)} \dots \langle q'', t'' | q', t' \rangle_{f,h} \Big|_{f=h=0}$$

1 Take $t' \rightarrow -\infty$, $t'' \rightarrow +\infty$ at fixed $\epsilon > 0$, and $\epsilon \rightarrow 0^+$ only at the end. The projection uses eigenstates of the source-free H , so f, h must vanish at early and late times. (Above, $e^{iE_n t}$ means $e^{iE_n t'}$.)

(7)

If we are interested in initial and final states other than position eigenstates, we will need to integrate them over. We are also taking $t' \rightarrow -\infty$, $t'' \rightarrow +\infty$

$$\langle 0|0 \rangle_{f,h} = \lim_{\substack{t' \rightarrow -\infty \\ t'' \rightarrow +\infty}} \int dq'' dq' \psi_0^*(q'') \langle q'', t'' | q', t' \rangle \psi_0(q')$$

$$\psi_0(q) = \langle q|0 \rangle, \quad \psi_0^* = \langle 0|q'' \rangle$$

↑ ground state wave function in q -space

let $|n\rangle$ denote an eigenstate of H with energy E_n . Suppose that $E_0 = 0$. If this is not true, we will shift H by such a constant.

$$\begin{aligned} |q', t'\rangle &= e^{iHt'} |q'\rangle = \sum_{n=0}^{\infty} e^{iHt'} |n\rangle \langle n|q'\rangle \\ &= \sum_{n=0}^{\infty} \psi_n^*(q') e^{iE_n t'} |n\rangle \end{aligned}$$

$\psi_n(q) = \langle q|n\rangle$ as the wave function of the n th eigenstate.

Now replace H with $(1-i\epsilon)H$, ϵ infinitesimal (positive) ①

Then take the limit $t' \rightarrow -\infty$

Every state except the ground state is multiplied by a ~~fixed constant~~ vanishing factor of $\lim_{t \rightarrow -\infty} e^{+\epsilon E_n t}$.
exponentially

2 A constant factor rather than a shift: $\int dq' \chi(q') \psi_0^*(q')$ and its q'' partner, nonzero if χ overlaps the ground state. Normalizing $\langle 0|0 \rangle_{f=h=0} = 1$ removes it.

Similarly, this also picks out the ground state for $t'' \rightarrow +\infty$

Further, we can multiply arbitrary functions $\chi(q')$ and integrate over, which will yield a constant shift that can be absorbed into $DqDq'$ [we then only need to care $\langle 0|0 \rangle_{f,h}$ Damping & project onto vacuum] ②

This implies that if we use

$$H \rightarrow (1-\epsilon)H, \text{ we can be } \textcircled{1}$$

cavalier about boundary conditions ~~as~~.

¹ Read $(1-\epsilon)H$ here and below, as on p. 8. $(1-\epsilon)H$ only rescales H and damps nothing; the $-\epsilon$ is what gives $e^{-\epsilon E_n |t|}$.

Any reasonable boundary conditions will result in good ground state as both the initial & final state

$$\langle 0|0 \rangle_{fh} = \int \mathcal{D}p \mathcal{D}q e^{i \int_{-\infty}^{\infty} dt (p\dot{q} - (1-\epsilon)H + f\dot{q} + h p)}$$

the above

$$\text{Suppose } H = H_0 + H_1$$

(note the connection with interaction picture, in contrast to Schrödinger Picture & Heisenberg Picture)

H_0 is solvable (e.g. free theory)

H_1 is a perturbation.

$$\langle 0|0 \rangle_{fh} = \int \mathcal{D}p \mathcal{D}q e^{i \int_{-\infty}^{\infty} dt (p\dot{q} - \underbrace{H_0(p,q)}_{(1-\epsilon)} - \underbrace{H_1(p,q)}_{(1-\epsilon)} + \overbrace{f\dot{q} + hp}^{f\dot{q} + hp})}$$

but we just for shorthand not write them out explicitly.

$$= e^{[-i \int_{-\infty}^{\infty} dt H_1(\frac{\delta}{\delta h(t)}, \frac{\delta}{\delta f(t)})]} \times \textcircled{2}$$

$$\int \mathcal{D}p \mathcal{D}q e^{i \int_{-\infty}^{\infty} dt (p\dot{q} - H_0(p,q) + f\dot{q} + h p)}$$

² The arguments need $1/i$: $H_1(\frac{1}{i} \frac{\delta}{\delta h(t)}, \frac{1}{i} \frac{\delta}{\delta f(t)})$, since these bring down p and q (p. 7); cf. p. 10.

we assume we know how to calculate this. (We shall see next a calculable example)

Now, if H only on q (not on p)
 and if we are only interested in time-ordered
 (simply)
 products of Q_s (not P_s), and if H is no
 more than quadratic on P and the term quadratic in P
 not dependent on Q , then we can simplify

the above equations to

$$\langle 0|0 \rangle_f = e^{i \int_{-\infty}^{+\infty} dt L_1(\frac{1}{i} \frac{\delta}{\delta f(t)})} \times$$

$$\int Dq \exp \left[i \int_{-\infty}^{+\infty} dt (L_0(\dot{q}, q) + f(q)) \right].$$

$$\text{Here } L_1(q) = -T_1(q).$$